Interview Questions for Back- end Developer

1. What is a backend developer?

- ANSWER:: A backend developer focuses on server-side logic, database interactions, and APIs to build and maintain the technology that powers the components enabling the user-facing side of a web application.

2. What is the difference between SQL and NoSQL databases?

- ANSWER:: SQL databases use structured query language for defining and manipulating data, are table-based, and have a predefined schema. NoSQL databases are more flexible, storing data in various formats like key-value pairs, documents, or graphs, and are schema-less.

3. Explain RESTful API.

- ANSWER:: A RESTful API is an API that adheres to the principles of REST (Representational State Transfer), using HTTP requests to GET, PUT, POST, and DELETE data, and is stateless, with each request containing all the information needed to process it.

4. What is an ORM?

- ANSWER:: An ORM (Object-Relational Mapping) is a programming technique that allows developers to interact with a database using the object-oriented paradigm of their programming language, abstracting away the SQL queries.

5. What are microservices?

- ANSWER:: Microservices are an architectural style where a large application is composed of small, independent services that communicate over a network. Each service is focused on a single business capability.

6. What is a primary key?

- ANSWER:: A primary key is a unique identifier for a record in a database table, ensuring that no two rows have the same primary key value.

7. What is a foreign key?

- ANSWER:: A foreign key is a field (or collection of fields) in one table that uniquely identifies a row in another table, creating a relationship between the two tables.

8. Explain the concept of database normalization.

- ANSWER:: Database normalization is the process of organizing the fields and tables of a relational database to minimize redundancy and dependency, typically through a series of normal forms.

9. What is a JOIN in SQL?

- ANSWER:: A JOIN is an SQL operation used to combine rows from two or more tables based on a related column between them. Common types include INNER JOIN, LEFT JOIN, RIGHT JOIN, and FULL JOIN.

10. What is ACID in database systems?

- ANSWER:: ACID stands for Atomicity, Consistency, Isolation, and Durability, which are properties that ensure reliable processing of database transactions.

11. What is indexing in databases?

- ANSWER:: Indexing is a database optimization technique that improves the speed of data retrieval operations on a database table at the cost of additional storage space and slower write operations.

12. What is a transaction in a database?

- ANSWER:: A transaction is a sequence of one or more SQL operations treated as a single unit of work. It ensures that either all operations are successfully completed, or none are, maintaining database integrity.

13. Explain database sharding.

- ANSWER:: Database sharding is a partitioning technique where a large database is split into smaller, more manageable pieces called shards, which can be distributed across multiple servers.

14. What is a stored procedure?

 - ANSWER:: A stored procedure is a precompiled collection of one or more SQL statements stored in the database, which can be executed as a single unit to perform a specific task.

15. What is the difference between a clustered and a non-clustered index?

 - ANSWER:: A clustered index determines the physical order of data in a table and is typically faster for retrieval. A non-clustered index maintains a logical order of data that differs from the physical order and includes a pointer to the physical data.

16. What is MVC architecture?

 - ANSWER:: MVC (Model-View-Controller) is an architectural pattern that separates an application into three main components: the Model (data), the View (user interface), and the Controller (business logic), promoting organized and modular code.

17. Explain the concept of middleware in web frameworks.

 - ANSWER:: Middleware refers to software that acts as an intermediary between different components of a web application, typically handling requests and responses. Examples include authentication, logging, and data transformation.

18. What is a REST API, and how does it differ from SOAP?

 - ANSWER:: REST API (Representational State Transfer) uses HTTP and supports various data formats, typically JSON, while SOAP (Simple Object Access Protocol) is a protocol that uses XML and has more rigid specifications.

19. What is a WebSocket?

- ANSWER:: WebSocket is a communication protocol providing full-duplex communication channels over a single, long-lived TCP connection, allowing real-time data transfer between client and server.

20. What is a daemon in the context of backend development?

- ANSWER:: A daemon is a background process that runs continuously and handles requests for services, often starting at boot time and without direct user interaction.

21. What is SQL injection, and how can you prevent it?

- ANSWER:: SQL injection is a code injection technique that exploits vulnerabilities in an application's software by inserting malicious SQL code. It can be prevented by using parameterized queries, prepared statements, and ORM frameworks.

22. What is cross-site scripting (XSS), and how can it be mitigated?

- ANSWER:: XSS is a security vulnerability where attackers inject malicious scripts into web pages viewed by others. It can be mitigated by sanitizing and escaping user input, using Content Security Policy (CSP), and validating output.

23. What is CSRF, and how can you defend against it?

- ANSWER:: CSRF (Cross-Site Request Forgery) is an attack where a malicious website causes a user's browser to perform an unwanted action on another site where the user is authenticated. Defenses include using anti-CSRF tokens, SameSite cookies, and validating the HTTP Referer header.

24. Explain the concept of OAuth.

- ANSWER:: OAuth is an open-standard authorization framework that allows third-party services to exchange information without exposing user credentials. It uses access tokens to grant limited access to user resources.

25. What is HTTPS, and why is it important?

- ANSWER:: HTTPS (HyperText Transfer Protocol Secure) is an extension of HTTP that uses SSL/TLS to encrypt data between the client and server, ensuring secure communication and protecting against eavesdropping and tampering.

26. What is load balancing, and why is it important?

- ANSWER:: Load balancing distributes incoming network traffic across multiple servers to ensure no single server becomes overwhelmed, improving application availability, reliability, and performance.

27. What is caching, and how does it improve performance?

- ANSWER:: Caching stores frequently accessed data in a temporary storage area for quick retrieval. It reduces latency and server load, improving the performance and scalability of an application.

28. What is a CDN, and how does it work?

- ANSWER:: A CDN (Content Delivery Network) is a network of distributed servers that deliver web content to users based on their geographic location, enhancing load times, reducing latency, and improving site availability.

29. Explain the concept of database replication.

- ANSWER:: Database replication involves copying and maintaining database objects in multiple database servers to ensure data availability, fault tolerance, and load balancing.

30. What are some common performance optimization techniques for backend applications?

- ANSWER:: Techniques include optimizing database queries, using caching mechanisms, implementing load balancing, using efficient algorithms and data structures, and performing asynchronous processing.

31. What is Docker, and why is it used?

- ANSWER:: Docker is a platform for developing, shipping, and running applications in containers. It provides a consistent environment for development, testing, and deployment, improving efficiency and scalability.

32. Explain the concept of Continuous Integration/Continuous Deployment (CI/CD).

- ANSWER:: CI/CD is a practice where developers integrate code into a shared repository frequently (CI) and deploy code to production quickly and safely (CD), using automation tools to streamline the process.

33. What is Kubernetes?

- ANSWER:: Kubernetes is an open-source container orchestration platform that automates the deployment, scaling, and management of containerized applications, ensuring they run efficiently in different environments.

34. What is Infrastructure as Code (IaC)?

- ANSWER:: IaC is the practice of managing and provisioning computing infrastructure using code and automation, allowing for version control, reproducibility, and scalability.

35. What is AWS Lambda?

- ANSWER:: AWS Lambda is a serverless compute service that lets you run code in response to events without provisioning or managing servers. You only pay for the compute time consumed.

36. What is event-driven architecture?

- ANSWER:: Event-driven architecture is a design paradigm where system components communicate and execute actions in response to events or messages, enabling decoupled and scalable systems.

37. What is the difference between monolithic and microservices architectures?

- ANSWER:: A monolithic architecture is a single, unified application, while a microservices architecture breaks down the application into small, independently deployable services that communicate over a network.

38. What is a message queue, and why is it used?

- ANSWER:: A message queue is a component that allows asynchronous communication between different parts of a system by storing messages until they can be processed. It helps decouple processes and improve

39. What is a reverse proxy, and how does it work?

- ANSWER:: A reverse proxy is a server that sits between client requests and backend servers, forwarding client requests to the appropriate backend server. It improves security, load balancing, and performance by hiding the identity of backend servers and caching content.

40. What is GraphQL, and how does it differ from REST?

- ANSWER:: GraphQL is a query language for APIs and a runtime for executing those queries by allowing clients to request specific data. Unlike REST, which has fixed endpoints and returns a fixed structure, GraphQL allows clients to query only the data they need.

41. What is a circuit breaker pattern, and when would you use it?

- ANSWER:: The circuit breaker pattern is a design pattern used to detect failures and encapsulate the logic of preventing a failure from constantly recurring during maintenance, temporary external service downtime, or unexpected system failures. It helps improve the resilience of a system.

42. Explain the concept of eventual consistency in distributed systems.

- ANSWER:: Eventual consistency is a consistency model used in distributed computing to achieve high availability. It guarantees that, given enough time, all replicas of a distributed system will eventually reach consistency, though they may be temporarily inconsistent.

43. What are Webhooks, and how do they work?

- ANSWER:: Webhooks are user-defined HTTP callbacks that are triggered by specific events. When an event occurs, the source site makes an HTTP request to a URL configured by the user, often used for real-time notifications.

44. What is a distributed tracing system, and why is it important?

- ANSWER:: A distributed tracing system tracks and visualizes requests as they move through different services in a distributed system. It is important for diagnosing performance issues, debugging errors, and understanding the flow of requests across services.

45. What are API gateways, and what role do they play in microservices architecture?

- ANSWER:: An API gateway acts as a single entry point for all client requests in a microservices architecture. It handles request routing, composition, and protocol translation, improving security, load balancing, and reducing complexity for clients.

46. What is CAP theorem, and how does it apply to distributed systems?

- ANSWER:: The CAP theorem states that in a distributed system, it is impossible to simultaneously achieve Consistency, Availability, and Partition Tolerance. Systems must choose to prioritize two out of the three based on their requirements.

47. What is a dead letter queue, and why would you use it?

- ANSWER:: A dead letter queue (DLQ) is a specialized message queue used to store messages that cannot be processed successfully. It helps in isolating and troubleshooting issues with message processing in a message queue system.

48. What is the purpose of using a Content Security Policy (CSP)?

- ANSWER:: CSP is a security feature that helps prevent a variety of attacks, including Cross-Site Scripting (XSS) and data injection attacks, by specifying which dynamic resources are allowed to load on a webpage.

49. What is service discovery, and why is it important in microservices?

- ANSWER:: Service discovery is the process of automatically detecting devices and services on a network. In microservices, it helps dynamically locate services, allowing for better scaling, load balancing, and resilience.

50. What is a read replica, and how does it help in database scalability?

- ANSWER:: A read replica is a copy of the primary database that is used to distribute read traffic and improve read performance. It helps in scaling the database by allowing read-heavy workloads to be handled by multiple replicas, reducing the load on the primary database.